Methotrexate: Mechanism of Action, Adverse Drug Reactions and Risk Reduction Measures

Drug Overview

Methotrexate is an antifolate antimetabolite drug that belongs to the class of Disease Modifying Anti-Rheumatic Drugs (DMARDs) and antineoplastic agents. It is widely used for the treatment of rheumatoid arthritis, psoriasis, acute lymphoblastic leukemia, lymphoma, osteosarcoma and several other malignancies. Methotrexate is included in the WHO Essential Medicines List and acts by interfering with folate metabolism, thereby inhibiting DNA synthesis and cellular proliferation.

Mechanism of Action

Methotrexate is a folic acid analogue that inhibits dihydrofolate reductase (DHFR). This enzyme converts dihydrofolate into tetrahydrofolate, which is required for the synthesis of purines and thymidylate necessary for DNA and RNA production. Inhibition of DHFR results in reduced DNA synthesis and decreased cell division, especially in rapidly proliferating cells. In autoimmune diseases, methotrexate inhibits AICAR transformylase, causing accumulation of AICAR. Increased AICAR promotes adenosine release, and adenosine suppresses inflammatory cytokines, reduces T-cell activation and decreases inflammation.

Reported Adverse Drug Reactions (ADRs)

Organ System	Reported Adverse Effects	Severity
Hepatic System	Elevated liver enzymes, hepatotoxicity, liver fibrosis, cirrhosis	Serious
Pulmonary System	Pneumonitis, pulmonary fibrosis, cough, dyspnea	Serious
Hematologic System	Anemia, leukopenia, thrombocytopenia, pancytopenia	Serious
Gastrointestinal System	Nausea, vomiting, diarrhea, stomatitis, oral ulcers	Moderate
Renal System	Renal impairment, increased serum creatinine	Moderate to Serious
Skin	Rash, photosensitivity, alopecia	Mild to Moderate
Nervous System	Headache, dizziness, fatigue	Mild
Reproductive System	Teratogenicity, fetal abnormalities, fetal death	Serious

Measures to Reduce Risks

Monitoring parameters include Complete Blood Count (CBC), Liver Function Tests (LFTs), Kidney Function Tests, Chest X-ray and pulmonary assessment where required. Methotrexate is contraindicated in pregnancy, breastfeeding, severe liver disease, severe kidney disease, active infections and severe bone marrow suppression. Dose adjustments may be required in patients with renal impairment, liver dysfunction and elderly patients. Folic acid supplementation, avoidance of alcohol, adequate hydration and regular follow-up help reduce toxicity.

Conclusion

Methotrexate is an effective drug used in both cancer therapy and autoimmune disorders. Although highly beneficial, it may produce serious hepatic, pulmonary and hematological adverse effects.

Appropriate monitoring and preventive measures can significantly improve patient safety and therapeutic outcomes.

References

1. WHO VigiAccess. Methotrexate ADR Reports.
2. DrugBank Online Database.
3. NCBI StatPearls: Methotrexate.
4. Katzung's Basic and Clinical Pharmacology.
5. Goodman & Gilman's The Pharmacological Basis of Therapeutics.